

User manual

Nano Board with STM32F303CCT6 MCU

Introduction

The Nano board allows users to easily develop applications with the STM32F303CCT6 microcontroller with the Arm Cortex-M4 32-bit core.

Based on STM32F303CCT6, it includes one MPU6050 (is a 3-axis gyroscope and a 3-axis accelerometer sensor), LEDs, push-buttons and a Micro-B USB port, and an onboard ST-Link V3 debugger.

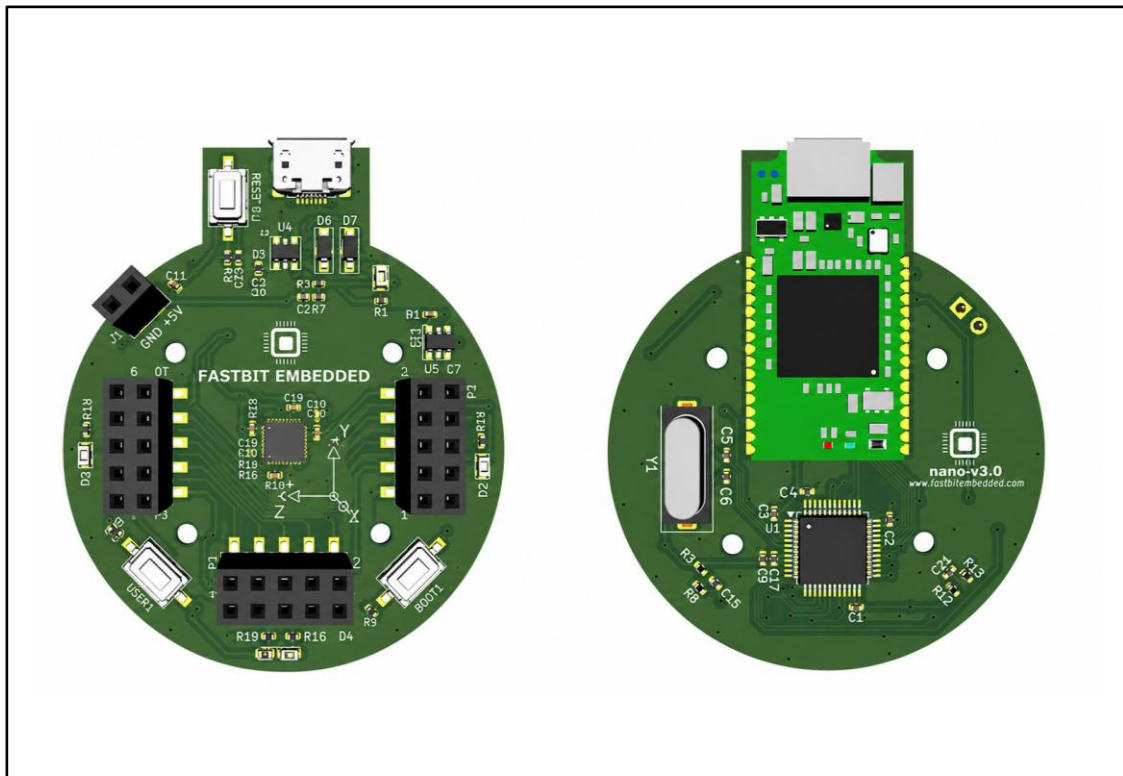
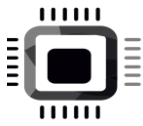
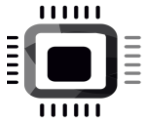


Figure 1: Nano



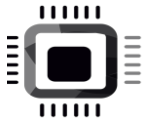
Contents

1. Features	7
2. List of tables	4
3. List of figures	5
4. Hardware and layout	8
5. Programming Nano Board	13



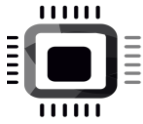
List of tables

Table 1.	Boot modes	10
Table 2.	HSE pins	11
Table 3.	UART1 pins	11



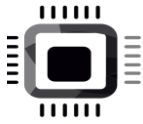
List of figures

Figure 1.	Nano	1
Figure 2.	Nano top layout	8
Figure 3.	Nano bottom layout	9



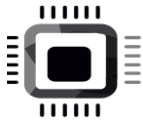
List of Block-diagram

Block 1:	ST-Link V3 (Virtual COM Port)	12
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List of abbreviation

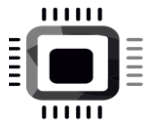
SWD	Serial Wire Debug
LQFP	Low Profile Quad Flat Package
HSE	High Speed External



Feature:

The Nano offers the following features:

- The STM32F303CCT6 microcontroller features a 32-bit Arm Cortex-M4 with FPU core (72 MHz max), 256-Kbyte Flash memory, and 48-Kbyte SRAM in a LQFP48 package.
- Three push buttons:
 - SW1(reset), SW2(user button), SW3(boot0).
- Four LEDs:
 - D1 (red) for 3.3 V power on / off.
 - Three user LEDs, D2 (blue), D3 (green), D4 (red).
- Board connectors:
 - J1: 5 V power output header (+5 V / GND).
 - J2 USB-B Micro.
 - 2.54 mm pitch extension header for 30 pins of LQFP48 I/Os for quick connection to prototyping board and easy probing.
- Power-supply options: ST-LINK or USB VBUS.



Hardware and layout

The Nano is designed around the STM32F303CCT6 microcontroller in a 48-pin LQFP package.

Figure 2 and Figure 3 help users to locate STM32F303CCT6 and its peripherals (MPU6050, push buttons, LEDs).

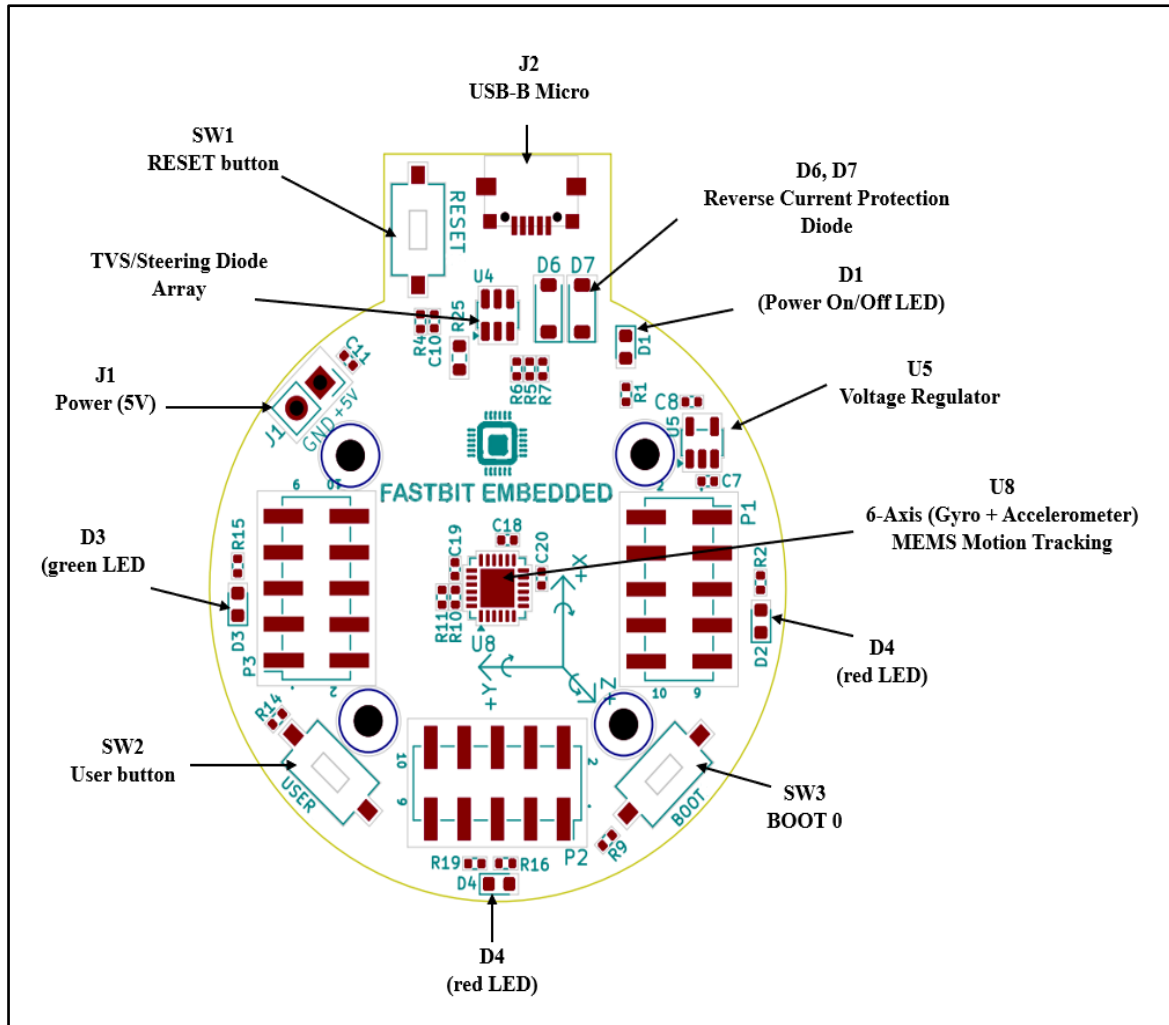


Figure 2: Nano top layout

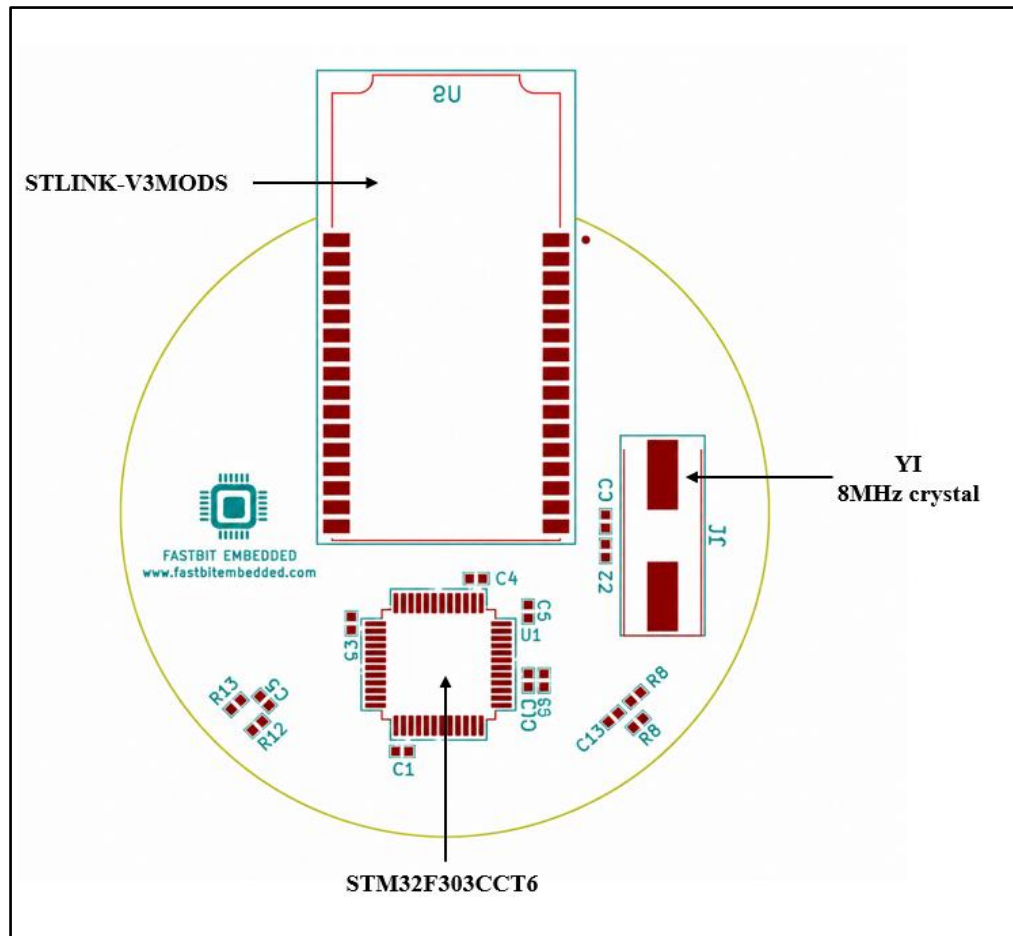
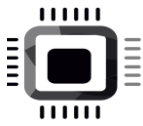
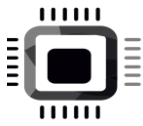


Figure 3: Nano bottom layout



Power supply and power selection

- The power supply is provided by the host PC through the Micro-B USB cable..
 - U5 regulates the 5V input from USB to provide a steady 3.3V output.
 - A regulated 5 V output is available on the J1 header for powering external circuitry.

Note: Don't connect external power supply to header sockets.

LEDs

- D1 PWR: red LED indicates that the board is powered.
- User D2: blue LED is a user LED connected to the I/O PA1 of the STM32F303CCT6.
- User D3: green LED is a user LED connected to the I/O PA2 of the STM32F303CCT6.
- User D4: red LED is a user LED connected to the I/O PA3 of the STM32F303CCT6.

Push buttons

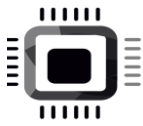
- SW1 RESET: Push button connected to NRST is used to RESET the STM32F303CCT6.
- SW2 USER: User button is connected to the I/O PA0 of the STM32F303CCT6.
- SW3 BOOT0: Push button connected to BOOT0 is used to toggle the boot mode of the STM32F303CCT6.

BOOT0	Boot Mode
0	Main flash memory
1	System memory

Table 1: Boot modes

Note: By default, the microcontroller runs the application code from the main flash memory. To change this behavior, use the BOOT0 button.

1. Press and hold the BOOT0 button and then press the reset button.
2. This action makes the microcontroller to run the built-in bootloader from the system memory (i.e., microcontroller enters the bootloader mode).

**6-Axis (Gyro + Accelerometer) MEMS Motion Tracking**

- MPU6050 sensor is a low power, low cost, and high-performance 6-axis (Gyro + Accelerometer).
- The MPU6050 devices combine a 3-axis gyroscope and a 3-axis accelerometer on the same silicon die, together with an onboard Digital Motion Processor (DMP), which processes complex 6-axis Motion Fusion algorithms.
- The STM32F303CCT6 microcontroller controls this sensor through the I2C interface.

OSC Clock

- **HSE: OSC 8 MHz clock supply**

Refers to an external oscillator running at 8MHz. It provides a higher-frequency clock source suitable for driving the core processing unit or other high-speed peripherals.

Pin Name	Pin Function
PF0	OSC_IN
PF1	OSC_OUT

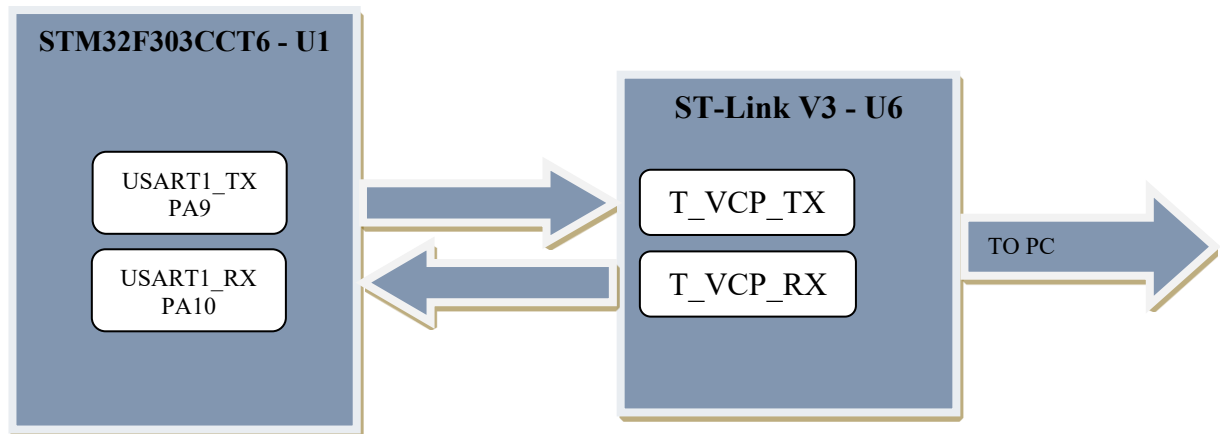
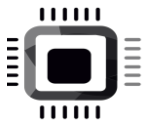
Table 2: HSE pins

On-board ST-Link V3 (Virtual COM Port)

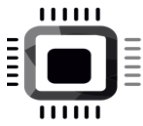
- The on-board ST-Link V3 has a built-in USB Virtual COM Port (VCP) for serial communication between a computer and the STM32F303CCT6. UART1Tx and UART1Rx serve as the transmit and receive pins (PA9 and PA10).

Pin Name	Pin Function
PA9	UART1_Tx
PA10	UART1_Rx

Table 3: UART1 pins



Block1: UART-USB Bridge



Programming Nano Board

1. Using boot loader (No Dedicated Debug Probe Required):

a. UART Bootloader Mode (via Onboard ST-Link V3 Virtual COM Port for UART Bridge)

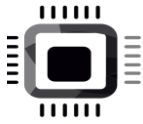
- Connect the board to the PC using the Micro-B USB cable. The onboard **ST-Link V3** provides a Virtual COM (UART) interface.
- Press and hold the **BOOT0** button, then press and release the **RESET** button.
- Release the **BOOT0** button.
- Open **STM32CubeProgrammer** and select **UART** as the connection interface.
- Select the appropriate COM port corresponding to the ST-Link V3 Virtual COM Port.
- Connect to the target MCU.
- Open the .elf, .hex, or .bin file and download it to the board.
- Press the **RESET** button to run the application.

Note: Debugging is not supported when programming through the UART bootloader.

b. USB DFU Mode

- Power the board via USB.
- Press and hold the **BOOT0** button, then press and release the **RESET** button.
- Release the **BOOT0** button.
- Open **STM32CubeProgrammer** and select **USB** as the connection interface.
- Connect to the board.
- Open the .elf, .hex, or .bin file and download it to the board.
- Press the **RESET** button to run the application.

Note: You cannot debug using this method. For information, follow the [link](#)



2. Using ST-Link:

- Connect the board to the PC using the Micro-B USB cable (J2). The onboard ST-Link V3 module handles programming and debugging, so no external programmer is required.
- Power the board via the Micro-B USB cable.
- Open STM32CubeIDE or any other compatible IDE.
- Configure the IDE to use ST-Link as the debugging interface.
- Build your project or open a .elf file.
- Use the IDE to download the program to the board.
- Use the IDE's "Debug" option to debug the program, or the "Run" option to execute the program without debugging.